



DOUGLAS COLLEGE

CSIS 4495 – 002

(Applied Research Project)

Title: EcoTots

(MERN Stack Website for the Exchange of Gently Used kid's Clothing)

Progress Report 4

Submitted To:

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1. Introduction

Since the last progress report, significant advancements have been made in refining the EcoTots platform. The primary focus has been on finalizing the homepage, implementing an improved search bar menu, and adding filters for condition (new or used) and gender (boys, girls, or unisex). Additionally, a "Trending Now" feature has been introduced to highlight popular listings dynamically. The AI chatbot is also being integrated to enhance user experience by assisting with search-related inquiries.

These enhancements aim to provide users with a more intuitive browsing experience while maintaining platform efficiency and security. This report outlines the latest progress, challenges encountered, and future plans for continued improvements.

2. Key Contributions

2.1 Homepage and About Page Completion

- Finalized the homepage UI to align with the previous design style.
- Improved responsiveness and visual consistency across different screen sizes.
- Added dynamic sections to highlight featured listings, new arrivals, and trending items.
- Integrated personalized recommendations based on user preferences.
- Updated About page (About.jsx) to include project details and team member information.

2.2 Search Bar and Filter Enhancements

- Implemented an intuitive search bar menu with autocomplete functionality.
- Optimized search queries to return more relevant results quickly.
- Developed category filters for better product discovery.
- Added filters for:
 - Condition: New or Used.
 - Gender: Boys, Girls, or Unisex.
- Ensured filter selections persist when navigating through search results.

2.3 Performance and Security Improvements

- Further optimized MongoDB queries to improve search speed and reduce server load.
- Implemented backend validations for search filters to prevent invalid data entries.
- Strengthened Firebase authentication mechanisms with improved session management.
- Conducted security audits on search queries to prevent potential vulnerabilities.

2.4 AI Chatbot Enhancements

- Improved chatbot UI for better user interaction.
- Integrated chatbot responses with the new search functionality, allowing users to receive assistance in finding items.
- Enhanced chatbot training with additional FAQs related to filtering and browsing.
- Developed chatbot-triggered messaging for seller inquiries (e.g., "Is this still available?").

2.5 "Trending Now" Feature

- Implemented a dynamic section on the homepage to showcase trending items.
- Items are ranked based on recent views, saves, and purchases.
- Ensured real-time updates using Firebase database tracking.

3. Challenges Faced and Solutions Implemented

Challenge	Solution Implemented
Slow search performance when applying multiple filters	Optimized database queries and indexing strategies
UI inconsistencies in homepage layout on mobile devices	Conducted extensive UI/UX testing and applied responsive design fixes
Users struggling with search filter usability	Improved filter layout and provided tooltips for guidance
Security concerns related to unauthorized query manipulation	Strengthened backend validations and implemented request monitoring

Ensuring chatbot provided relevant search assistance	Trained chatbot with additional queries related to search and filtering
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4. Repo Check-in Summary

Backend Updates:

- **Controllers:** Updated search.controller.js and listing.controller.js to improve search functionality.
- **Models:** Enhanced listing.model.js to include optimized filtering mechanisms.
- **Routes:** Improved API endpoints for handling search and filtering queries securely.
- **Middleware:** Strengthened validation checks for search input parameters.

Frontend Updates:

- **Pages:** Finalized Home.jsx, updated About.jsx with project and team information, and improved Search.jsx for better user experience.
- **Components:** Added and refined Search.jsx.
- **Redux Store:** Updated searchSlice.js to manage search and filtering states efficiently.

Miscellaneous Updates:

- Uploaded latest user feedback reports to the repository.
- Implemented additional security fixes to prevent unauthorized data manipulation.

5. Work Log Summary

Date	Hours	Description of Work Done
Mar 17	2.0	Finalized homepage layout and UI improvements
Mar 18	1.5	Improved mobile responsiveness for homepage
Mar 19	2.0	Developed and tested search bar autocomplete functionality
Mar 20	2.5	Implemented category and condition filters in the search bar

Mar 21	2.0	Optimized search query performance and indexing
Mar 22	1.5	Conducted UI/UX testing and fixed filter usability issues
Mar 23	2.5	Integrated chatbot assistance for search-related inquiries
Mar 24	2.0	Added "Trending Now" feature to homepage
Mar 25	2.0	Improved chatbot AI responses for better user interactions
Mar 26	2.0	Enhanced chatbot integration with user search queries
Mar 27	1.5	Fixed minor UI issues on the homepage
Mar 28	2.5	Improved chatbot logic for personalized recommendations
Mar 29	2.0	Conducted additional performance testing for search filters
Mar 30	2.5	Strengthened chatbot security and request validation

6. Future Plans

- Further enhance search and filter functionalities with additional criteria like price range and brand.
- Continue optimizing search performance for large datasets.
- Expand AI chatbot capabilities to offer more personalized recommendations.
- Implement analytics tracking to analyze search behavior and improve user experience.
- Conduct A/B testing on homepage design to determine the most effective layout.
- Strengthen fraud detection measures for marketplace transactions.

7. Conclusion

This phase of development has successfully completed key enhancements to the homepage, search bar, filtering features, AI chatbot, and About page, significantly improving the platform's usability and efficiency. With continued security improvements and AI chatbot integration, EcoTots is progressing toward a more user-friendly and secure marketplace experience. Future work will focus on refining search functionalities,

expanding chatbot assistance, and enhancing fraud detection mechanisms to build a more robust and trusted platform.

This report highlights the ongoing contributions and their impact on creating a scalable and efficient platform for exchanging children's clothing.